Project name: The hospital Booking System

Team name: Aura members

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PROBLEM

Patients currently rely too much on hospital staff, members, calls, and even in-person visits to check the availability of the hospital rooms and book appropriate doctors. A long queue, unpredictable waiting hours, and unavailability of doctors after a long wait can result in the disappointment of patients (customers). It is becoming more serious these days as hospitals are overcrowded than they used to. To solve these major problems, we design a system where patients can check the availability of rooms and make appointments with doctors effortlessly via the system online.

GATE 1 — REACHABLE USERS

1. นพ.คณพัฒน์ พรพิพิธ - Family Medicine Physician, Phramongkutklao Hospital – 24 Aug 2026
2. นางสาวอัมรา ตาแกว้ - Housewife – 24 Aug 2026
3. นางพิศ ปัญญาวรรณ – Housewife – 24 Aug 2026
4. นางฐาณมาศ กิติภัทย์พิบูลย์ – Housewife – 24 Aug 2026
5. นางสาว พิมพ์พิชชา คุณมาศ – Collegian – 24 Aug 2026

GATE 2 — MVP SHAPE

<u>9 planned functional requirements:</u>

FR-1: View Doctor Information and Availability

The system shall allow all the users (patients) to view all the active doctors, their specializations and degrees, departments, qualifications from various majors, profiles, and available appointment schedules within the website.

FR-2: Book an Appointment and Receive a Token

The system shall allow patients to book available appointments and receive an automatically generated token number immediately after a successful booking.

FR-3: View Personal Token and Appointment Status

The system shall allow all patients to view their personal token numbers, appointment details with accurate date and time, and latest appointment statuses.

FR-4: Cancel or Reschedule an Appointment and Arrange a Follow-up

The system shall allow patients who have already booked to cancel or reschedule their eligible appointments and arrange follow-up appointments after completing a visit.

FR-5: Manage Doctor Information and Schedules

The system shall allow the website Admins to add, update, and deactivate doctor profiles when they have left the hospital and manage all their schedules.

FR-6: View admin Dashboard, Appointments, and Token Records

The system shall allow Admins to monitor all the hospital statistics, appointment information, and token records.

FR-7: Manage Patient Account Access

The system shall allow Admins to view patient accounts, ban accounts, and restore patient access.

FR-8: Register, sign in, and Manage Patient Profile

The system shall allow the patients (users) to register, sign in, and manage their personal profile information.

FR-9: Receive and Manage In-App Notifications

The system shall allow patients and Admins to receive and manage notifications about appointments, account activities, and system events.

Which FR carries a real rule or decision (not plain CRUD)?

FR 3- Book an appointment

The system shall allow registered and authenticated patients to book an appointment by selecting an available doctor, date, and time slot. Before confirming the booking, the system shall verify whether the chosen time slot is still available or not. If the slot is unavailable or has already been taken by another patient, the system shall reject the booking and ask the patient to select another available time for them. If the booking is successful, the system shall save it and immediately generate a unique token number for that patient.

FR-3: A time slot that is already taken cannot be booked by another patient.

What data has to persist between sessions?

The system needs to persist the following data:

- Patient account information (personal and login information)
- Doctor information and their specializations (degrees and their medical background)
- Doctor schedules and available appointment slots
- Appointment information, including booking dates, times, and appointment status
- Generated token numbers related with appointments
- Token status (Waiting, Called, In Operation, Completed, and Cancelled)
- Appointment history which includes cancelled and completed appointments

This persistent data is required so that patients and admins can continue to access and manage appointment records, doctor schedules, and queue information after logging out, closing the browser, or starting a new session.

GATE 3 — DATA ACCESS

Our Hospital OPD Booking System requires most of the detailed information about the doctors, their schedules, appointments, and token numbers.

For the MVP, we do not ask patients to access their unnecessary medical records which is not related to current appointments. We will be using publicly available information from real

hospitals and use it as a reference to create samples for doctors, patients, schedules, appointments, and token numbers.

We learn how real hospitals OPD booking System works. Depending on what we learn, we are going to design our own booking processes and use sample data to develop the working system.

PROCESS & TEAM

Incremental/Iterative model (or Agile) — Because a hospital booking system has many requirements that change over time and needs feedback and review from people including doctors, admin staff and patients. It helps lower risk by delivering working parts of the system.